

Milestone 4 – Deployment & Final Documentation

AI TraceFinder: Scanner Image Verification System

Project Title

AI TraceFinder – Deep Learning–Based Scanner Image Verification System

Milestone

Milestone 4: Deployment & Final Documentation

1. Introduction

With the rapid growth of digital documentation, ensuring the authenticity and source verification of scanned documents has become a critical requirement in digital forensics. Scanner devices introduce unique noise patterns and artifacts that can be learned and identified using deep learning techniques.

AI TraceFinder is a scanner identification system designed to analyze scanned document images and determine their originating scanner device. Milestone 4 focuses on deploying the trained Convolutional Neural Network (CNN) model into a user-friendly application, enabling real-world usability and validation.

This milestone bridges the gap between model development and practical deployment by providing an end-to-end inference pipeline.

2. Objectives

The primary objectives of Milestone 4 are:

- Deploy the trained CNN model for real-time inference
- Develop a web-based user interface for image upload
- Display predicted scanner label with confidence score
- Log all predictions for traceability and auditing
- Test the deployed system using sample images
- Provide complete documentation with screenshots

3. System Architecture

The deployed system follows a modular and sequential architecture designed for simplicity and efficiency.

Architecture Flow

1. User uploads a scanned document image

2. Image is preprocessed (resizing and normalization)
3. Preprocessed image is passed to the trained CNN model
4. Model generates prediction and confidence score
5. Results are displayed on the UI
6. Prediction details are saved to a CSV log file

4. Deployment Process

The deployment was performed locally using **Streamlit**, which provides an interactive and lightweight framework for machine learning applications.

Deployment Steps

Step 1: Install Dependencies

```
pip install -r requirements.txt
```

Step 2: Run the Application

```
python -m streamlit run app.py
```

Step 3: Access the Application

<http://localhost:8501>

Once launched, the application loads the trained model and waits for user input.

5. Model Description

- Model Type: Convolutional Neural Network (CNN)
- Training Completed In: Previous Milestones
- Input Image Size: **224 × 224 × 3**
- Output: Scanner class prediction
- Model Format: .h5
- Model File Name: tracefinder_model.h5

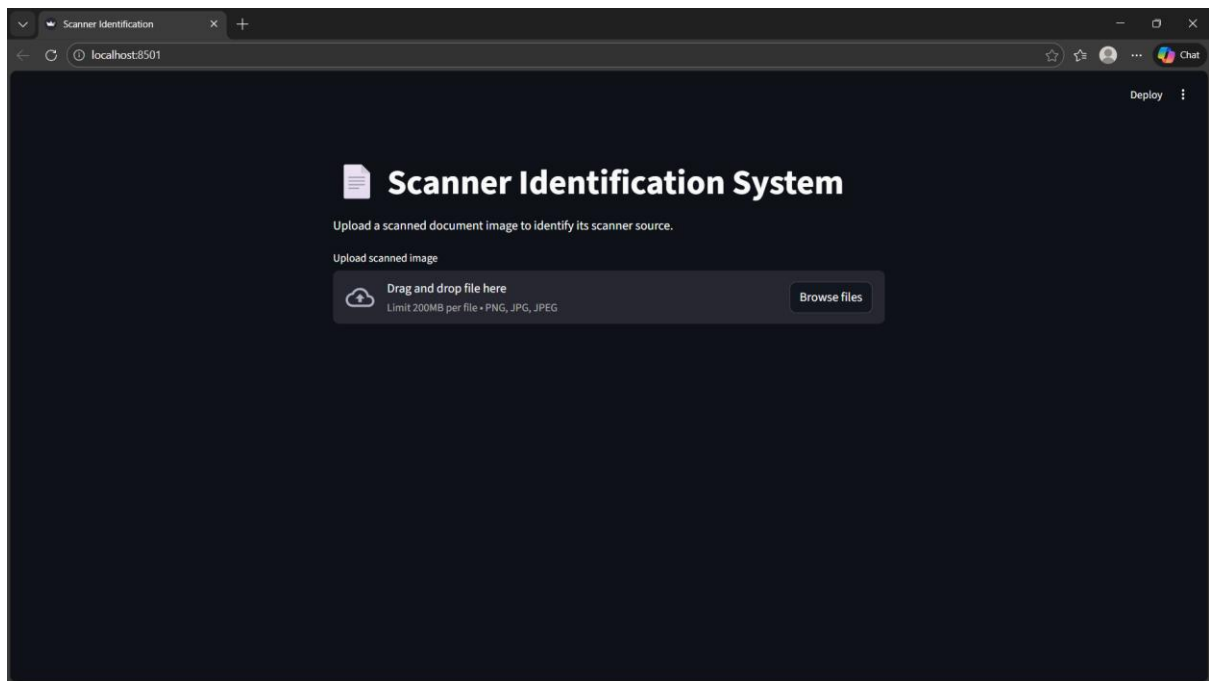
The model learns scanner-specific noise patterns embedded in scanned document images.

6. User Interface Description

The Streamlit-based interface provides:

- Image upload option
- Uploaded image preview
- Prediction label display
- Confidence score visualization
- Automatic prediction logging

The interface is designed to be minimal, intuitive, and suitable for forensic analysts.



7. Testing and Validation

Test Images Used

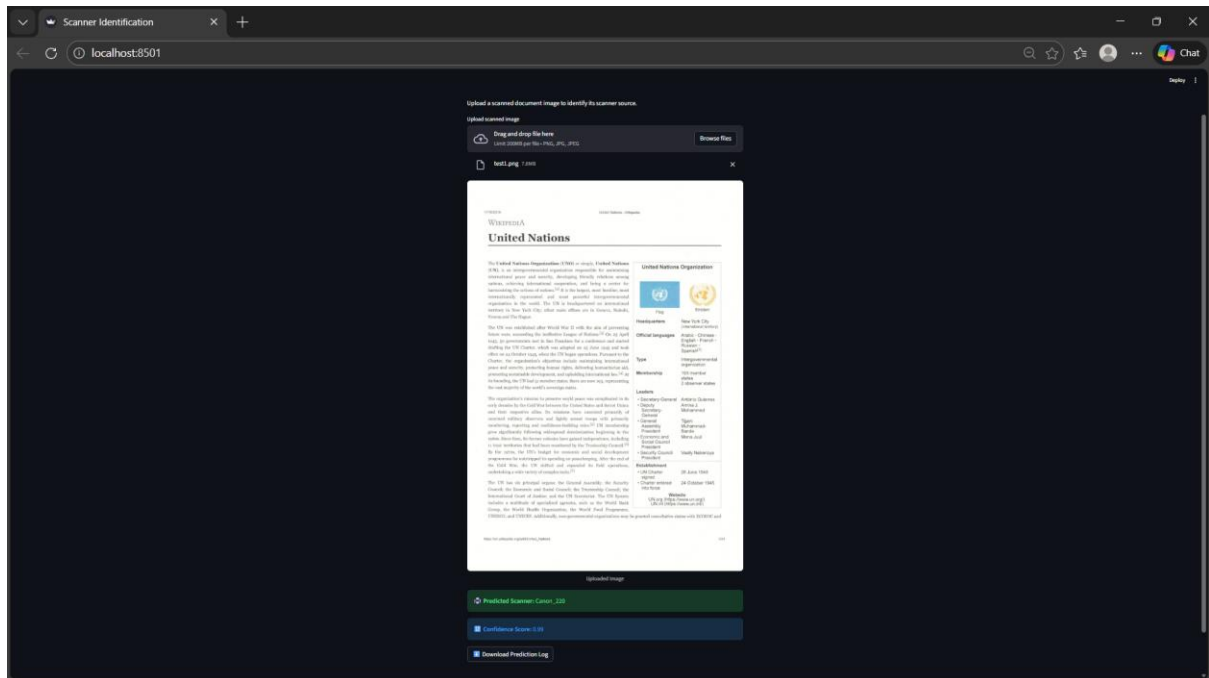
- test1.png
- test2.png
- test3.png

Testing Outcome

- All images were successfully processed
- Predictions were generated without errors
- Confidence scores were displayed correctly

- Logs were saved automatically

This confirms the stability and correctness of the deployed pipeline.



8. Prediction Logging

To ensure transparency and traceability, each prediction is logged in a CSV file.

Log File

- File Name: prediction_log.csv

Stored Fields

- Timestamp
- Image Name
- Predicted Label
- Confidence Score

The log file can be downloaded for reporting or further forensic analysis.

9. Results and Screenshots

The result_screenshots directory contains:

- Streamlit application UI
- Uploaded image preview
- Prediction and confidence output

These screenshots validate the correct functioning of the deployed system.

10. Limitations

Despite successful deployment, certain limitations remain:

- Large model size may exceed GitHub storage limits
- Deployment is local and not cloud-hosted
- Performance depends on scan quality and resolution
- Model generalization may be limited to trained scanner devices

11. Future Enhancements

Possible future improvements include:

- Cloud deployment using AWS / GCP
- REST API integration using FastAPI
- Multi-scanner and brand-level classification
- Explainability using Grad-CAM visualization
- Secure user authentication and access control

12. Conclusion

Milestone 4 successfully completes the AI TraceFinder pipeline by deploying the trained CNN model into a fully functional application. The system allows scanned image uploads, generates predictions with confidence scores, logs results, and provides an intuitive interface for users.

This milestone demonstrates real-world applicability of scanner identification techniques and lays the foundation for advanced forensic document verification systems.